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Notation

$\mathbb{N} = \{1, 2, 3, \dots\}$	set of natural numbers (positive integers)
$\mathbb{Z}_n = \{0, 1, \dots, n-1\}$	ring of residue classes modulo n
$\mathbb{Z}, \mathbb{Q}, \mathbb{R}, \mathbb{C}$	sets of integer, rational, real and complex numbers
$\mathbb{R}_+$ and $\mathbb{R}_{\geq 0}$	sets of positive and non-negative real numbers
$\mathbb{R}^n$	n -dimensional real space
$\mathbb{F}_q$	finite field of size $q = p^k$
$]a, b[$ and $[a, b]$	open and closed interval from a to b
$ x $	absolute value of x
$\#M$ or $ M $	cardinality of a set M
$M \times N$	Cartesian product of two sets or graphs M and N
$M \cup N, M \cap N, M \setminus N$	union, intersection and difference of two sets M and N

1. Polygons from Polygons

1. a) Given a rectangle $ABCD$, let X be a point inside the rectangle. Show that it is possible to choose three of the four segments XA, XB, XC and XD which can be sides of a triangle.
b) Find or estimate possible angles, perimeters and areas of such triangles.
2. a) A quadrilateral $ABCD$ is given in the real plane. Find the locus of all points X such that among the four segments XA, XB, XC and XD , there are three which can be sides of a triangle.

b) Find or estimate possible angles, perimeters and areas of the triangles that can be obtained in such a way.

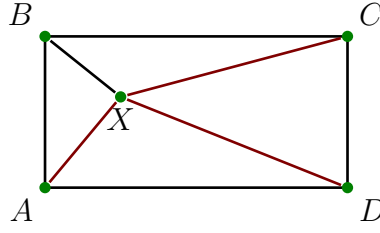


FIGURE 1. A point in a rectangle.

3. a) Let $n, k \geq 3$ be positive integers. A regular n -gon P is given in the real plane. Find the locus of all points X such that among the segments from X to the vertices of P , there are k segments which are sides of a convex k -gon.

b) Find or estimate possible angles, perimeters and areas of the k -gons that can be obtained in such a way.

4. Consider the previous question when P is a non-regular convex n -gon.

5. Consider the question when P is a non-convex n -gon.

6. Generalize and investigate the problem in dimension 3.

7. Suggest and study additional directions research.

2. A Fractional Inequality

1. a) Find the least real constant C such that the following inequality holds for any positive real numbers $x, y, z \geq C$:

$$\frac{x^6}{y^4 + z^3} + \frac{y^6}{z^4 + x^3} + \frac{z^6}{x^4 + y^3} \geq \left(\frac{x + y + z}{3} \right)^2.$$

b) Find all positive real numbers x, y, z for which the above inequality holds. When is equality attained?

2. Let $n \geq 4$ be a positive integer. Consider the set T_n of all tuples $(x_1, x_2, \dots, x_n) \in \mathbb{R}_+^n$ for which the following inequality holds:

$$\frac{x_1^6}{x_2^4 + x_3^3} + \frac{x_2^6}{x_3^4 + x_4^3} + \dots + \frac{x_n^6}{x_1^4 + x_2^3} \geq \left(\frac{x_1 + x_2 + \dots + x_n}{n} \right)^2.$$

Find the set T_n , and study its properties.

3. Let n and a, b, c, d be positive integers. Find all $x_1, x_2, \dots, x_n \in \mathbb{R}_+$ for which the following inequality holds:

$$\frac{x_1^a}{x_2^b + x_3^c} + \frac{x_2^a}{x_3^b + x_4^c} + \dots + \frac{x_n^a}{x_1^b + x_2^c} \geq \left(\frac{x_1 + x_2 + \dots + x_n}{n} \right)^d.$$

When is equality attained?

4. On the left hand side of the inequality, replace the cyclic sum $\sum_{cyc} \frac{x_i^a}{x_{i+1}^b + x_{i+2}^c}$ by another cyclic fractional sum, for example $\sum_{cyc} \frac{x_i^a}{x_{i+1}^b + x_{i+3}^c}$, and solve it.

5. On the right hand side of the inequality, replace the arithmetic mean $\frac{x_1 + x_2 + \dots + x_n}{n}$ by another mean and solve it.

6. Consider further generalizations and investigate them.
7. Suggest and study additional directions research.

3. Coprime Blocks

1. Prove that for all sufficiently large positive integers n , there is a block of n consecutive positive integers containing no integer relatively prime to the others.
2. For any positive integers c , and $d \geq 2$ and for all large n , there are n consecutive terms of the arithmetic progression $(c + kd)_{k \in \mathbb{Z}}$ such that no integer is relatively prime to each of the others.
3. Let $P(x) = ax^2 + bx + c$ be a polynomial with integer coefficients, prove that for all large k , there is a positive integer n such that in the list $P(n+1), P(n+2), \dots, P(n+k)$ no integer is relatively prime to each of the others.
4. Deduce the same result for the cubic polynomial $P(x) = ax^3 + bx^2 + cx + d$ with integer coefficients.
5. Try to generalize the result for the polynomials of degree $d \geq 4$ with integer coefficients.
6. In the previous questions, try to find the minimal integer C such that the statement is true for all $n \geq C$.
7. Consider an analogous question for the sequence $a_n = c^n + d$, where $\gcd(c, d) = 1$.
8. Suggest and study additional directions of research.

4. Inequalities for Divisors

Denote by $\sigma(n)$ the sum of all positive divisors of a positive integer n . For example, $\sigma(6) = 1 + 2 + 3 + 6 = 12$.

1. a) Show that for all positive integers n and $\lambda = 2$ holds the following inequality:

$$\sigma(1) \cdot \sigma(2) \cdot \dots \cdot \sigma(n) < \lambda^n \cdot n!.$$

b) Try to find the infimum of all λ that satisfy the inequality above for all sufficiently large enough positive integers n .

2. a) Prove that for any positive integer m there exists a constant c_m such that for any positive integers n , we have the inequality

$$S(n, m) := \frac{\sigma(1^m)}{1^m} + \frac{\sigma(2^m)}{2^m} + \dots + \frac{\sigma(n^m)}{n^m} \leq c_m n.$$

For a given m , define $c_*(m)$ to be the infimum of all such c_m .

- b) For which m is the following property satisfied $\lim_{n \rightarrow \infty} (S(n, m) - c_*(m)n) = 0$?

c) Let P be a polynomial with integral coefficients of degree d . Try to find sufficient conditions on P that guarantee existence of a constant c_P such that for any $n \in \mathbb{N}$,

$$\frac{\sigma(P(1))}{P(1)} + \frac{\sigma(P(2))}{P(2)} + \dots + \frac{\sigma(P(n))}{P(n)} \leq c_P \cdot n.$$

Try to estimate such a constant c_P by means of coefficients or roots of the polynomial P .

3. a) Prove that for any real number $x > 2$ there exists a real number $M(x)$ such that for any $n \in \mathbb{Z}^+$, we have

$$\frac{\sigma(1)}{1^x} + \frac{\sigma(2)}{2^x} + \dots + \frac{\sigma(n)}{n^x} \leq M(x),$$

while for any $x \leq 2$ there exists no such $M(x)$.

b) Is it true that there exist some constants C and D for which the inequality

$$a_n := \frac{\sigma(1)}{1^2} + \frac{\sigma(2)}{2^2} + \dots + \frac{\sigma(n)}{n^2} \leq C \ln n + D$$

is valid for all $n \in \mathbb{N}$?

c) Could you find a continuous function $f: \mathbb{R} \rightarrow \mathbb{R}$ such that $\lim_{n \rightarrow \infty} \frac{a_n}{f(n)} = 1$?

4. Suggest and study additional directions of research.

5. Polynomial Sets

For a given polynomial $P \in \mathbb{Q}[x]$ and set $L \subseteq \mathbb{R}$ let us define the set

$$P(L) = \{P(r) \mid r \in L\}.$$

We will be interested in investigating properties of $P(L)$ and their relations to L and P .

1. a) Show that there exists such polynomials P and Q with rational coefficients such that $P(\mathbb{Q}) = Q(\mathbb{Q})$ but $P \neq Q$.

b) Does there exist the set L for which the equality $P(L) = Q(L)$ for some polynomials $P, Q \in \mathbb{Q}[x]$ implies $P = Q$?

2. Is it true that $P(L)$ uniquely determines the degree of P if

a) $L = \mathbb{Z}$; b) $L = \mathbb{Q}$; c) $L = \{x^3 - 1 \mid x \in \mathbb{Z}\}$?

3. Try to describe all pairs $(P, Q) \in \mathbb{Q}[x]^2$ such that $P(\mathbb{Q}) = Q(\mathbb{Q})$.

4. Suppose, for some polynomial P with integer coefficients and some prime number $q \in \mathbb{N}$ holds the following property: for any positive integer $n \neq 10^{10}$ we have $q^n \in P(\mathbb{N})$. Prove or disprove that $q^{10^{10}} \in P(\mathbb{N})$.

5. Find all polynomials $P \in \mathbb{Q}[x]$ that satisfy the following property: for all $a \in P(\mathbb{Z})$ we have $g(a) \in P(\mathbb{Z})$, where $g(x) = x^2 + 1$.

6. Consider the previous question for other polynomials $g \in \mathbb{Q}[x]$.

7. Suggest and study additional directions research.

6. Boundaries of Graphs

1. Let $G = (V, E)$ be a connected undirected graphs. For any nonempty subset of vertices $A \subseteq V$ define the set

$$\partial A = \{v \in V \setminus A \mid \exists u \in A : (u, v) \in E\},$$

which is called the *vertex boundary* of A . For any graph G we define the quantity

$$h(G) = \min\{|\partial A|/|A| \mid A \subset V, 0 < |A| \leq |V|/2\},$$

which is called the *vertex isoperimetric number* of graph G .

Let $(G_n)_{n \in \mathbb{N}}$ be a sequence of connected undirected graphs. Find magnitude of $h(G_n)$, that is, find best possible constants α, β such that inequalities $c_1 n^\alpha \leq h(G_n) \leq c_2 n^\beta$ hold for some positive real constants c_1, c_2 and any sufficiently large positive integer n , for the following graphs:

- a) $G_n = P_n$ is a path on n vertices;
- b) $G_n = K_n$ is a complete graph on n vertices;
- c) $G_n = P_n \times P_n$ is a grid graph on the plane;
- d) $G_n = Q_n$ is the graph of the hypercube in $\mathbb{R}^n$.

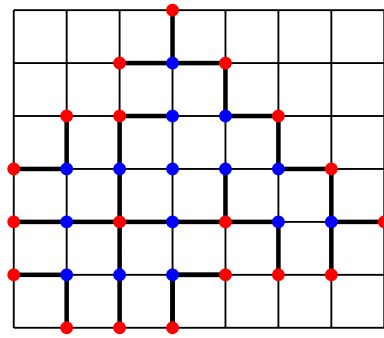


FIGURE 2. The vertex boundary ∂A (red vertices) of a subset A (blue vertices) of the vertices of a grid graph $P_6 \times P_7$.

2. The *permutohedron* A_n is a polytope in $\mathbb{R}^n$, the vertices of which are formed by permuting the coordinates of the vector $(1, 2, \dots, n)$, and two vertices $(a_1, \dots, a_n)$ and $(b_1, \dots, b_n)$ are connected if and only if one is obtained from the other by interchanging the numbers i and $i + 1$ for some $1 \leq i \leq n - 1$. Let $G_n = A_n$ be the graph of the permutohedron of order n . Find magnitude of $h(G_n)$.

3. Let $\alpha \in (0, 1]$ be a real number. Define the generalized vertex isoperimetric number of graph G in the following way:

$$h_\alpha(G) = \min\{|\partial A|/|A|^\alpha \mid A \subset V, 0 < |A| \leq |V|/2\}.$$

Compute exact value or estimate $h_\alpha(G)$ for the graphs listed above.

4. Suggest and investigate your own directions of this problem. For example, it might be interesting to obtain efficient algorithms to compute vertex isoperimetric numbers for some families of graphs.

7. Isosceles Tetrahedra

Let a tetrahedron $ABCD$ be given. We will say that tetrahedron $ABCD$ is *isosceles* if all its faces are congruent triangles. A *median* of a tetrahedron is a segment connecting the vertex of the tetrahedron and the centroid of the opposite face. A *bisector* of a tetrahedron is a segment enclosed inside the tetrahedron and being the bisector of the trihedral angle formed by the edges of the tetrahedron coming out from one vertex. In this problem, we will consider barycentric coordinates with respect to the tetrahedron $ABCD$.

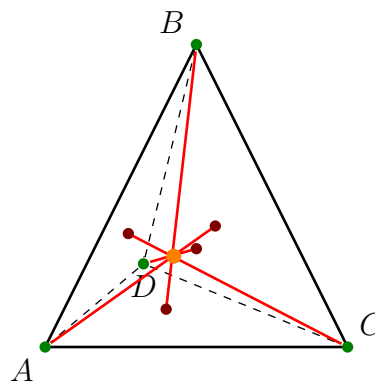


FIGURE 3. Medians (red lines) in a tetrahedron $ABCD$.

1. a) Prove that all medians of the tetrahedron intersect at one point.

- b) Determine in what ratio the medians of the tetrahedron are divided by the intersection point of the medians.
- c) Prove that the intersection point of the medians has barycentric coordinates $(1 : 1 : 1 : 1)$.
2. a) Prove that all the bisectors of the tetrahedron intersect at one point.
- b) Determine in what ratio the bisectors of the tetrahedron are divided by the intersection point of the bisectors.
- c) Prove that the intersection point of the bisectors has barycentric coordinates $(S_A : S_B : S_C : S_D)$, where S_A, S_B, S_C, S_D are the areas of the faces of the tetrahedron opposite to the vertices A, B, C, D respectively.
3. Is it true that tetrahedron $ABCD$ is necessarily isosceles if:
- All the heights of the tetrahedron are equal?
 - All the medians of the tetrahedron are equal?
 - All the bisectors of the tetrahedron are equal?
4. Is a tetrahedron necessarily isosceles if three of its four heights are equal? Consider a similar problem for three medians and three bisectors.
5. Suppose that two heights in a tetrahedron are equal. Is it true that the faces onto which these heights are dropped are equal? Investigate a similar problem for two medians and two bisectors.
6. Let k be a real number. Consider the point O_k inside the tetrahedron that has barycentric coordinates $(S_A^k : S_B^k : S_C^k : S_D^k)$. We call a k -bisector a segment enclosed inside the tetrahedron that passes through one of the vertices of the tetrahedron and the point O_k . Investigate previous questions for the k -bisectors of the tetrahedron.
7. Define an analogue of a tetrahedron in the n -dimensional Euclidean space $\mathbb{R}^n$, its heights, medians and bisectors, and investigate similar questions.
8. Suggest and study additional directions research.

8. Chromatically Critical Graphs

Given a finite simple graph $G = (V, E)$, its *chromatic number* $\chi(G)$ is the smallest positive integer k for which there exists a function $f : V \rightarrow \{1, 2, \dots, k\}$ such that $f(v_1) \neq f(v_2)$ whenever $v_1 v_2 \in E$. If G has $\chi(G) = k$, then we call it k -critical if $\chi(H) < k$ for every subgraph H of G , other than G itself.

Let $k \geq 2$ be an integer, let $n \geq 1$ be the number of vertices of G , and let $e \geq 1$ be the number of edges of G .

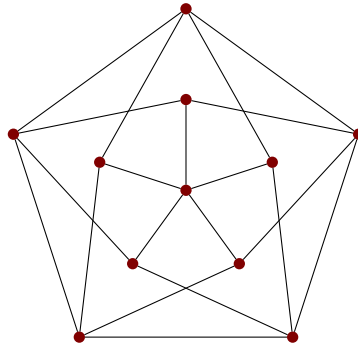


FIGURE 4. An example of a 4-critical graph.

- Describe all k -critical graphs for each $k \in \{2, 3, 4\}$.

2. a) For every $k \geq 3$ prove that every k -critical graph is 2-connected.
 b) Is there a k such that every k -critical graph is 3-connected?
3. In terms of k , what is the smallest possible number of vertices of a k -critical graph? Describe all graphs which attain the minimum.
4. a) Prove that a k -critical graph on n vertices has at least $\left(\frac{k-1}{2} + \frac{k-3}{2(k^2-3)}\right)n$ edges.
 b) Could you improve this bound?
5. a) For any integer N construct a 6-critical graph $G = (V, E)$ with $N \leq e \leq 5n$.
 b) Could you improve the constant 5? That is, does there exist a real number $c < 5$ such that for any integer N there is a 6-critical graph with $N \leq e \leq cn$? If yes, then find the smallest possible such c (or the infimum of such c , in case the minimum does not exist).
6. a) For each integer $n \geq 1000$ decide whether there is a 6-critical graph on n vertices and $e \geq \left\lceil \frac{n^2}{4} \right\rceil$ edges.
 b) What is the largest possible constant c (or the supremum of such c , if a maximum does not exist) such that for infinitely many n there is a 6-critical graph on n vertices and $e \geq cn^2$ edges?
 c) What happens if we require ‘sufficiently large n ’ instead of just ‘infinitely many n ’ in b)?
7. Consider questions 5 and 6 for k -critical graphs for an integer $k > 6$.
8. Consider questions 1–7 when we replace ‘every subgraph H ’ by ‘every induced subgraph H ’ in the definition in the beginning.
9. Suggest and study additional directions of research.

9. Irreducible Polynomials

A polynomial $f(x)$ with integer coefficients, the leading of which is equal to 1, is said to be *irreducible over $\mathbb{Z}$* if there do not exist non-constant polynomials $g(x)$ and $h(x)$ with integer coefficients such that $f(x) = g(x)h(x)$.

Let $n \geq 2$ and $a \neq 0$ be integers. We are interested in some polynomials of the form $f_{n,a}(x) = x^n - x + a$.

1. a) Suppose n is prime and $|a| \geq 3$. Prove that if n does not divide a or $n = \pm a$, then $f_{n,a}$ is irreducible over $\mathbb{Z}$.
 b) Find, in terms of a where $|a| \leq 2$, all (not necessarily prime) values of n for which $f_{n,a}$ is irreducible over $\mathbb{Z}$.
2. Now we consider $a = n$. Suppose that $f_{n,n} = gh$ with $g, h \in \mathbb{Z}[x]$ being non-constant integer polynomials, and denote $g(0) = b$, $h(0) = c$.
 a) Show that $b > 1$ and $c > 1$.
 b) Show that $b^{n-1} \equiv 1 \pmod{c}$ and $c^{n-1} \equiv 1 \pmod{b}$. Additionally, if c is prime, then show that $b^{b-1} \equiv 1 \pmod{c}$.
 c) Prove that if $n - 1$ is prime, then $f_{n,n}$ is irreducible over $\mathbb{Z}$.
 d) Prove that $f_{n,n}$ is irreducible over $\mathbb{Z}$ for as many n as you can.
 e) Is there actually a value of n for which $f_{n,n}$ is reducible?
3. Investigate irreducibility over $\mathbb{Z}$ of $f_{n,a}$ for $a = -n$, $2n$ or n^2 .
4. Investigate irreducibility of polynomials $f_{n,a}$ over finite fields.
5. Suggest and study additional directions of research.

10. Functions, Groups and Graphs

For a function f , we define $f^{(0)}(x) \equiv x$, and let $f^{(n+1)}(x) = f(f^{(n)}(x))$ be iterations of f .

Let p and q be primes. Denote by $A(p, q, 1)$ the set of all non-identity bijections f of the set $\mathbb{Z}_p = \{0, 1, \dots, p-1\}$ to itself such that

$$f(x+y) \equiv f(x) + f(y) \pmod{p} \quad \text{for all } x, y \in \mathbb{Z}_p$$

and $f^{(q)} = f^{(0)}$.

1. a) Given p and q find the number of elements in $A(p, q, 1)$. In particular, prove that there are infinitely many pairs (p, q) with non-empty $A(p, q, 1)$.

b) Let $\Gamma_1 = \Gamma_1(p, q, 1)$ be a graph on $A(p, q, 1)$ such that f and g are joined by the edge if and only if $f(g(x)) = g(f(x))$ for all $x \in \mathbb{Z}_p$ and $\Gamma_2 = \Gamma_2(p, q, 1)$ be the complement of Γ_1 . Find or estimate the number of edges in Γ_1 and Γ_2 .

c) Is Γ_i connected for $i \in \{1, 2\}$?

d) Find or estimate the maximal diameter of a connected component of Γ_i for $i \in \{1, 2\}$.

e) Is Γ_i regular for $i \in \{1, 2\}$?

2. In the set $\mathbb{Z}_p^2 = \{(a, b) \mid a, b \in \mathbb{Z}_p\}$, consider the operation $(a, b) \oplus (c, d) = (a+c \pmod{p}, b+d \pmod{p})$. Denote by $A(p, q, 2)$ the set of all non-identity bijections f of the set $\mathbb{Z}_p^2$ to itself such that

$$f(x \oplus y) = f(x) \oplus f(y) \quad \text{for all } x, y \in \mathbb{Z}_p^2$$

and $f^{(q)} = f^{(0)}$. Solve analogues of question 1 in this case.

3. How do answers to questions 1 and 2 change if we remove the condition $f^{(q)} = f^{(0)}$ in the definition of our functions?

4. Suggest generalizations and study them.

Email address: oc@itym.org

URL: <http://www.itym.org/>